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Histopathological characteristics of secondary enucleation after iodide-125 plaque radiotherapy in Chinese patients with uveal melanoma

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ABSTRACT

Purpose To explore the histopathological characteristics in eyes with uveal melanoma (UM) that had secondary enucleation after iodine-125 plaque radiotherapy (PRT).

Methods This study included 243 patients with unilateral UM treated at Beijing Tongren Hospital from 2007 to 2020. Of these, 119 initially received PRT, followed by enucleation 2 months–9 years later. We retrospectively examined the histopathological features of enucleated ocular tissues to investigate microscopic differences based on treatment methods and reasons for secondary enucleation.

Results Compared with 124 primary enucleations, eyes after PRT were more likely to show tumour diffuse distribution; had more tumour cell necrosis and more inflammation. In addition, eyes after PRT had a higher frequency of sclera invasion, optic nerve invasion by the tumour and neovascularisation of the iris. Histopathologically, 55 eyes (46%) were removed due to local control failure. Compared with eyes enucleated because of severe radiotherapy complications, eyes enucleated for local control failure had tumours invading the optic disc and ciliary body more frequently, pathological mitosis and the proportion of tumour diffuse distribution were higher. Eyes with radiotherapy complications had more tumour tissue necrosis and more neovascularisation of the iris.

Conclusions By observing the pathological features, we regrouped the patients who underwent secondary enucleation after PRT. These differences in histopathology may represent tissue effects of radiotherapy radiation or features related to tumour progression and growth. Tumour insensitivity and progression after PRT clearly increased the risk of metastasis-related death. Some pathological microscopic features may be important prognostic indicators of patients with UM.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. Histopathology holds an irreplaceable significance in the field of oncology.

WHAT THIS STUDY ADDS

⇒ This study retrospectively examined the histopathological features of enucleated ocular tissues to investigate microscopic differences based on treatment methods and reasons for secondary enucleation.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study provides a new pathological perspective for understanding the biological effects of plaque radiotherapy on UM. Microscopic features, such as pathological mitotic activity, may be an important prognostic indicator for patients with UM.

INTRODUCTION

Uveal melanoma (UM), the most common primary intraocular malignancy in adults, exhibits varying annual incidence rates across regions, ranging from 0.42 to 0.64 per million in East Asia[1] to approximately 4–5 per million in Western countries.[2] Treatment decision-making requires a comprehensive and individualised approach.[3] Plaque radiotherapy (PRT) and enucleation are the most common treatments. PRT is the preferred treatment method for small- and medium-sized UM and has induced tumour regression and preserved the globes in most cases.[4]

Unfortunately, in some cases, local treatment failure and severe complications of the radiotherapy, such as keratitis, iris neovascularisation, neovascular glaucoma (NVG) and scleral necrosis,[4] may necessitate enucleation of the affected eye. In addition, the randomised, multicentre clinical trials conducted by the collaborative ocular melanoma study group showed no difference in long-term survival between patients treated with PRT and enucleation for medium-sized tumours.[5]

Although clinical and molecular risk factors for UM progression and complications have been extensively investigated, histopathological evaluations following PRT remain scarce. Histopathology plays a fundamental role in oncological diagnosis, treatment assessment and prognosis prediction, serving as a critical bridge between basic research and clinical practice. Some studies have evaluated the histopathological features of proton beam irradiation,[6] cobalt-60 PRT[7] and helium ion irradiation[8] in a few cases of UM. But these studies are mainly based on Caucasians. Patients with UM of different ethnicities may have different clinical characteristics, complications and outcomes.[9] For

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instance, while tumour recurrence was identified as the main indication for secondary enucleation in Western cohorts,[10] our previous research indicated that NVG is the leading cause among Chinese patients. Non-responsive or recurrent tumours further increase the risk of metastasis and mortality.[11]

Therefore, elucidating the histopathological characteristics of UM after brachytherapy in Chinese patients is clinically imperative. In this study, we analysed the histopathological findings and long-term prognosis of 119 enucleated eyes from Chinese patients with UM previously treated with iodide-125 PRT.

METHODS

Study population

This is a single-centre retrospective cohort study that was conducted in accordance with the Declaration of Helsinki and was approved by the Medical Ethics Committee of Beijing Tongren Hospital, Capital Medical University. Written informed consent was obtained from all participants.

We reviewed data from patients with UM who presented at Beijing Tongren Hospital between March 2011 and November 2021. The study included 119 patients who underwent secondary enucleation following iodine-125 (I^{125}) PRT and had completed regular follow-up. A control group of 124 patients who received primary enucleation was selected, matched by sex, age and tumour size.

Exclusion criteria included: (1) receipt of any UM treatment other than PRT or enucleation—such as laser therapy, transpupillary thermotherapy, photodynamic therapy (PDT), proton beam therapy (PBT) or local resection or (2) incomplete medical records or poor-quality imaging that precluded reliable evaluation.

All patients were initially diagnosed with UM using comprehensive ophthalmic examinations, including fundus microscopy, colour Doppler ultrasonography (CDI), MRI, fluorescein fundus angiography and indocyanine green angiography. Diagnosis was subsequently confirmed through histopathological examination of enucleated specimens. Clinical and pathological diagnoses were validated by two senior ophthalmic oncologists (WW and XX) at our centre.

Clinical data

Baseline clinical characteristics collected prior to treatment included gender, age, laterality, tumour shape, site of tumour origin, largest basal diameter (LBD), tumour thickness and presence of iris neovascularisation. Tumour morphology was categorised as mushroom shaped, flat, hemispherical or irregular. Measurements of LBD and tumour thickness were derived from CDI.

PRT treatment

All the PRT surgeries were performed by the same surgeon (YL). The radiation time was decided based on tumour height,[12] and the plaque was 4 mm larger than the tumour LBD. During the operation, the anterior margin of the tumour base was located by indirect ophthalmoscopy, and iodide-125 plaque was sutured to the sclera. The radiation dose to be delivered to the tumour apex was 100 Gy.

Follow-up

Patients were followed using fundus microscopy and CDI commencing 1-month post-treatment, with subsequent evaluations at 3 months, 6 months and 1 year after plaque removal. Positron emission tomography–computed tomography were performed when metastasis was suspected. In cases of extraocular extension, orbital MRI typically demonstrated a mass with high T1-weighted and low T2-weighted signal intensity.

Follow-up was conducted semiannually to annually, with the timeline initiated from the initial treatment. Documented outcomes included radiotherapy complications, reasons and timing for secondary enucleation—such as keratitis, NVG, scleral necrosis or intractable ocular pain. NVG was defined as intraocular pressure ≥ 21 mm Hg on at least three separate measurements accompanied by iris neovascularisation. Scleral necrosis was identified as scleral thinning and enhanced transparency in the absence of tumour growth.

Patient outcomes, including metastasis and death due to melanoma or other causes, were recorded. Secondary enucleation following I^{125} brachytherapy was attributed to two principal causes: local treatment failure or severe radiotherapy-related complications. Treatment failure, assessed by ophthalmologists during follow-up, was defined as tumour growth, local recurrence or non-response according to established criteria.[13] Local recurrence was confirmed if previously regressed tumours exhibited an increase in basal diameter > 0.5 mm or height increase ≥ 15%. Non-response was defined as the absence of regression or continued tumour growth within the first year after PRT.[13]

Histopathological features

The following histopathological parameters were assessed: tumour location; involvement of the ciliary body or optic disc; extraocular extension; iris neovascularisation; pathological tumour type (based on the Callender classification[14]: spindle, epithelioid, mixed or necrotic) and the extent of necrosis, pigmentation, inflammatory infiltration and mitotic activity. Mitotic figures were counted per 40 high-power fields (HPFs; 40 × objective and 10 × ocular), with only unequivocal mitoses included.[15] Some typical histopathological images are shown in figure 1. Tumour location was categorised as posterior to the equator, equatorial, between the equator and ora serrata, ciliary body or diffuse. Necrosis was graded as: none, <10%, 10%–50%, 50%–90%, >90% or complete (100%). Inflammatory infiltration was scored as: 1 (absent/rare), 2 (moderate) or 3 (dense/diffuse). Pigmentation was graded as: 1 (minimal/none), 2 (moderate) or 3 (heavy, requiring bleaching for evaluation).[8]

Specimens with viable tumour cells were further analysed to identify the patterns of treatment failure. Local recurrences were subclassified as: marginal growth (residual cells at the periphery leading to flat extension), apical growth (residual cells at the summit leading to vitreous growth), global growth (multidirectional regrowth), extrascleral growth (extrascleral/orbital invasion) or distant growth (new intraocular tumour ≥ 5 mm from the original site).[16]

Patient demographic and clinical information were concealed, and the histopathological assessments were performed independently by two ophthalmic pathologists. If the two observations did not agree, further judgement was made by a third senior pathologist.

Statistical analysis

Categorical variables were compared between the I^{125} plaque failure group and primary enucleation group using the χ^2 test. The plaque failure cohort was further subdivided into local control failure and radiotherapy complication groups, which were also compared using the χ^2 test.

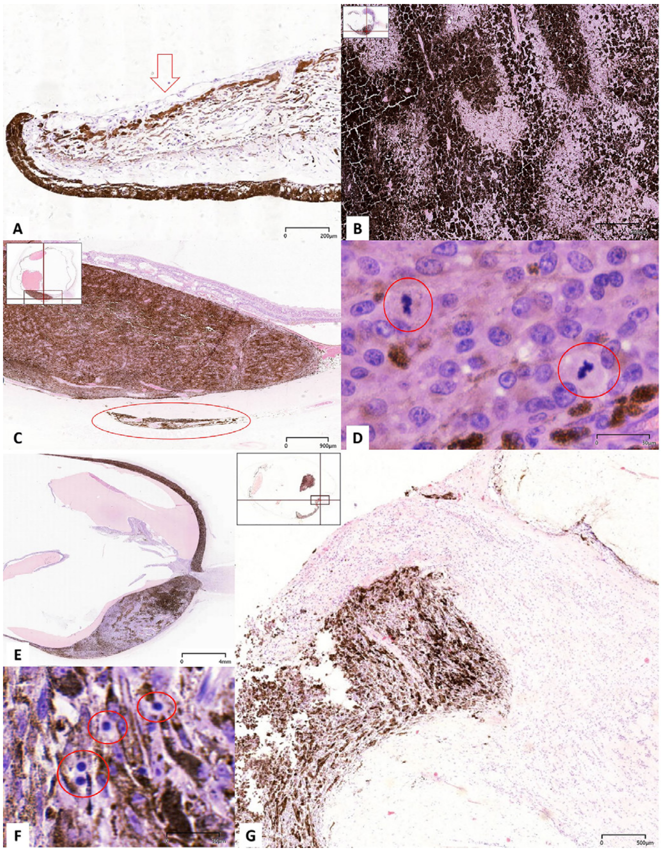

Figure 1 Typical histopathological images (H&E staining). (A) Iris neovascularisation and iris curl at the pupillary margin. (B) The tumour tissue is completely necrotic and cannot identify the pathological type. (C) Tumour cells and pigment invade scleral tissue. (D) Pathological mitotic image (red circle). (E) Tumour body diffuse distribution. (F) A few inflammatory cells (red circle) in the tumour. (G) Tumour cells invade the optic nerve.

Discrete variables were binarised, while continuous variables (age, diameter and thickness) were standardised. Survival outcomes were analysed using Kaplan–Meier curves for univariate analysis and Cox proportional hazards models for multivariate analysis. Variables significant in univariate analyses were included in the Cox regression model to assess the independent predictors of metastasis and metastasis-related mortality. All analyses were performed using SPSS 25.0 (IBM, NY, USA), with a p value$<$0.05 considered statistically significant.

RESULTS

Baseline clinical data and pathological features

Patients' clinical and histopathological features are summarised in table 1. After matching, there was no statistical difference in demographic information and tumour size between the two groups. Preoperative CDI assessment showed no significant difference in tumour shape between the two groups but a significant difference in tumour location (p=0.003). The tumours in the PRT group were basically located in the choroid behind the ciliary body, and there were no diffuse distribution cases, which was also consistent with the clinical indication of PRT.

As observed from the pathological tissue sections, most tumours in the PRT group did not involve the optic disc (56%) and ciliary body (77%), and most tumours did not have the episcleral extension (56%) and optic nerve invasion (81%). Iris neovascularisation was observed in 78 eyes (66%). Cell necrosis was observed in 80% of the tumours, of which 24 eyes could not identify the tumour cell type due to extensive tumour necrosis. Of these tumours, 36% had moderate or diffuse inflammatory cell infiltration, 83% had moderate or diffuse pigmentation and 42% had moderate or diffuse intratumoral blood vessels. 58 tumours (49%) showed more than one pathological mitosis per 40 HPFs.

The histopathological characteristics of melanomas in the PRT group were compared with primary enucleation (table 1). Our statistical results show that eyes after PRT were more likely to show tumour diffuse distribution (p=0.10). Compared with the PRT group, primary enucleation lesions showed less inflammation (p=0.04) and less tumour cell necrosis (p$<$0.001). No cell necrosis in 97 cases (78%). There was a smaller proportion of tumour invasion of the sclera (p=0.001) and optic nerve (p=0.04) in the primary enucleation group. Iris neovascularisation was also observed in 11 cases (9%) of the primary enucleation group.

Histopathological observation of 243 UM cases showed that the eyeballs and tumour tissues of secondary enucleation after the PRT group were pathologically different from those of primary enucleation. Compared with 124 primary enucleations, 119 eyes after PRT were more likely to show tumour diffuse distribution; tumours after PRT had more tumour cell necrosis, more broken tissue and more inflammation. In addition, compared with primary enucleations, eyes after PRT had a higher frequency of sclera invasion, optic nerve invasion by the tumour and neovascularisation of the iris.

Histopathological data of secondary enucleation group

According to the clinical medical record, 55 eyes were removed for local control failure after the PRT, and 52 eyes were enucleated because of severe radiotherapy complications without a reported concern of tumour growth (50 cases of NVG and 2 cases of scleral necrosis melting). In addition, three patients underwent enucleation due to the patient's request, and nine patients complained of eye pain, but no iris neovascularisation was observed clinically.

The clinical and histopathological characteristics of tumour local control failure eyes were compared with eyes having radiotherapy complications after PRT (online supplemental table 1). There was no statistical difference in demographic information and tumour size between the two groups.

In the local control failure group, a certain proportion (13%) of tumours diffuse distribution. 56% of tumours involved the optic disc, and 29% involved the ciliary body. Most tumours did not have episcleral extension (58%) and optic nerve invasion (84%). Iris neovascularisation was observed in 28 eyes (51%). Cell necrosis was observed in 80% of the tumours. Moderate or diffuse inflammatory pigmentation was observed in 76% of the tumours. 32 tumours (58%) showed more than one pathological mitotic per 40 HPFs.

As for eyes having radiotherapy complications, the vast majority (89%) of tumours were located in the equator and the choroid posterior to the equator. Most tumours did not involve the optic disc (71%) and ciliary body (85%), and most tumours did not have the episcleral extension (58%) and optic nerve invasion (83%). Iris neovascularisation was observed in 45 eyes (87%). Cell necrosis was observed in 81% of the tumours. And in 18 eyes (35%), the tumour cell type could not be determined due to extensive tumour necrosis. 89% of the tumours showed moderate or diffuse pigmentation. 32 tumours (62%) showed no pathological mitosis per 40 HPFs.

Table 1 Tumour characteristics of iodide-125 PRT eyes compared with primary enucleation eyes

Characteristics	Secondary enucleation after PRT (n=119)		%	Primary enucleation (n=124)		%	P
Clinical characteristics							
Age	50.2±10.7			51.1±12.7			0.46
Gender	Male	59		Male	56		0.29
	Female	60		Female	68		
Laterality	Right	60		Right	59		0.53
	Left	59		Left	60		
LBD	10.8±3.4			11.8±3.5			0.64
Tumour thickness	6.4±2.9			7.4±2.8			0.63
Tumour location	Posterior to equator	83	72%	Posterior to equator	61	49%	0.003
	At equator	25	22%	At equator	39	32%	
	Anterior to equator	7	6%	Anterior to equator	15	12%	
	Ciliary body	0	0	Ciliary body	0	0	
	Diffuse distribution	0	0	Diffuse distribution	5	4%	
Tumour shape	Mushroom	42	35%	Mushroom	51	41%	0.16
	Hemisphere	42	35%	Hemisphere	49	40%	
	Flat	11	9%	Flat	10	8%	
	Irregular	18	15%	Irregular	8	7%	
Histopathological characteristics							
Tumour location	Posterior to equator	74	62%	Posterior to equator	61	49%	0.01
	At equator	24	20%	At equator	39	32%	
	Anterior to equator	5	4%	Anterior to equator	15	12%	
	Ciliary body	2	2%	Ciliary body	0	0	
	Diffuse distribution	10	8%	Diffuse distribution	5	4%	
Involved the optic disc	None	66	56%	None	92	74%	0.003
	Yes	50	42%	Yes	31	25%	
Involved the ciliary body	None	92	77%	None	83	67%	0.05
	Yes	24	20%	Yes	39	32%	
Scleral extension	None	67	56%	None	99	80%	0.001
	Scleral ducts	40	34%	Scleral ducts	18	15%	
	Episcleral extension	7	6%	Episcleral extension	4	3%	
Optic nerve invasion	None	96	81%	None	112	90%	0.04
	Anterior to lamina cribrosa	14	12%	Anterior to lamina cribrosa	11	9%	
	Beyond the lamina cribrosa	5	4%	Beyond the lamina cribrosa	0	0	
Iris neovascularisation	None	32	27%	None	111	90%	< 0.001
	Yes	78	66%	Yes	11	9%	
Pathological types	Spindle	14	12%	Spindle	38	31%	< 0.001
	Epithelioid	49	41%	Epithelioid	42	34%	
	Mixed	32	27%	Mixed	39	32%	
	Necrotic	24	20%	Necrotic	3	2%	
Tumour cell necrosis	None	24	20%	None	97	78%	< 0.001
	< 10%	12	10%	< 10%	18	15%	
	10%–50%	27	23%	10%–50%	4	3%	
	50%–90%	19	16%	50%–90%	3	2%	
	> 90%	14	12%	> 90%	1	1%	
	100%	21	18%	100%	0	0	
Inflammation	None or light	76	64%	None or light	102	82%	0.004
	Moderate	26	22%	Moderate	14	11%	
	Diffuse	16	13%	Diffuse	7	6%	
Pigmentation	None or light	19	16%	None or light	29	23%	0.11
	Moderate	47	40%	Moderate	55	44%	
	Diffuse	52	44%	Diffuse	39	32%	
Tumour vessels prominent	None or light	68	57%	None or light	81	65%	0.38
	Moderate	36	30%	Moderate	32	26%	
	Diffuse	14	12%	Diffuse	10	8%	

Continued

Table 1 Continued

Characteristics	Secondary enucleation after PRT (n=119)		%	Primary enucleation (n=124)		%	P
Mitoses	0	60	50%	0	80	65%	0.07
	1	41	35%	1	28	23%	
	≥2	17	14%	≥2	15	12%	

LBD, largest basal diameter; PRT, plaque radiotherapy.

The two groups were compared according to pathological tissue observation. Eyes enucleated for local control failure had a greater proportion of presumed viable tumour in the lesion (p=0.002) and pathological mitosis (p=0.02), had a higher frequency of invasion of the optic disc (p=0.001) and ciliary body (p=0.03) and a higher proportion of tumour spread along the eye wall (p=0.04).

Eyes enucleated because of severe radiotherapy complications had more tumour tissue necrosis (p=0.002) and more neovascularisation of the iris ($p<0.001$).

Clinical and pathological reasons for secondary enucleation

Only according to the clinically reported reasons for the second enucleation after PRT may errors be caused by the authenticity of clinical medical records and the subjectivity of clinical observations. We regrouped patients with secondary ocular enucleation after PRT treatment based on the pathological characteristics. The consistency of the two groups was 83% (99/119). Of the 55 patients with clinical failure of local control, 49 had tumour recurrence or continued growth confirmed by pathology.

Eyes from secondary enucleation after PRT were further analysed microscopically to confirm the evidence of the type of tumour growth. There were 23 cases of marginal growth, 18 cases of global growth, 4 cases of apical growth, 3 cases of extrascleral growth and no distant growth. Some marginal growth tumours may present as tumour tissue diffuse distribution. Compared with preoperative CDI and MRI examination, it is generally considered to be a pathological manifestation of radiation insensitivity and tumour progression. Figure 2 illustrates the global growth of two tumours. Figure 2A shows the tumour body initially originated in the anterior to equator, and histopathological examination after enucleation revealed that the tumour tissue invaded the anterior chamber angle, iris and even the corneal endothelium, without obvious necrosis of the tumour cells. Figure 2B shows that this tumour originated in the choroid posterior to the equator and persistent growth of the tumour in all dimensions after PRT treatment.

In addition, the proportion of iris neovascularisation in the PRT group was as high as 66% (78/119), of which 12 cases had no clinical observation of neovascularisation but were confirmed by pathology.

Follow-up and prognosis data

The median follow-up time of the primary enucleation group was 45.18 (range 3.13–154.6) months. 119 patients treated with secondary enucleation after PRT were followed up for a median of 56.63 (range 3.13–166.37) months. The median time from primary PRT to secondary enucleation was 22.67 (range 2.03–110.77) months. In the metastatic setting, liver was the most common site (n=26), followed by lung (n=5), bone (n=4), brain (n=1), mediastinal lymph nodes (n=1) and multiple systemic metastases (n=3).

The 1-, 3-, 5- and 10-year survival rates for patients treated with PRT were 99.12% (95% CI 94% to 100%), 90.05% (95% CI 83% to 94%), 76.45% (95% CI 66% to 84%) and 41.93% (95% CI 27% to 57%), respectively, giving a median survival of 9.35 years (95% CI 6.15 to 12.55). The 1-, 3- and 5-year survival rates for the primary enucleation groups (not treated with PRT) were 96.73% (95% CI 90% to 99%), 81.15% (95% CI 71% to 88%) and 67.95% (95% CI 56% to 78%), respectively (figure 3A).

We further displayed a Kaplan–Meier survival plot to analyse metastasis-free survival in patients with secondary resection after PRT due to local control failure and radiotherapy complications (figure 3B). Patients with radiotherapy complications had a lower risk of metastasis-related death than those with local control failure ($p<0.001$, HR=3.12). The 3-, 5- and 10-year survival rates for patients with radiotherapy complications were 98.39% (95% CI 89% to 100%), 91.33% (95% CI 78% to 97%) and 57.72% (95% CI 35% to 75%), respectively. The 1-, 3-, 5- and 10-year survival rates for patients with local control failure were 98.08% (95% CI 87% to 100%), 79.99% (95% CI 66% to 89%), 58.49% (95% CI 43% to 71%) and 33.11% (95% CI 17% to 50%), respectively. All in all, the long-term survival rate of patients with local control failure after PRT was the lowest, giving a median survival of 6.28 years (95% CI 5.27 to 7.28).

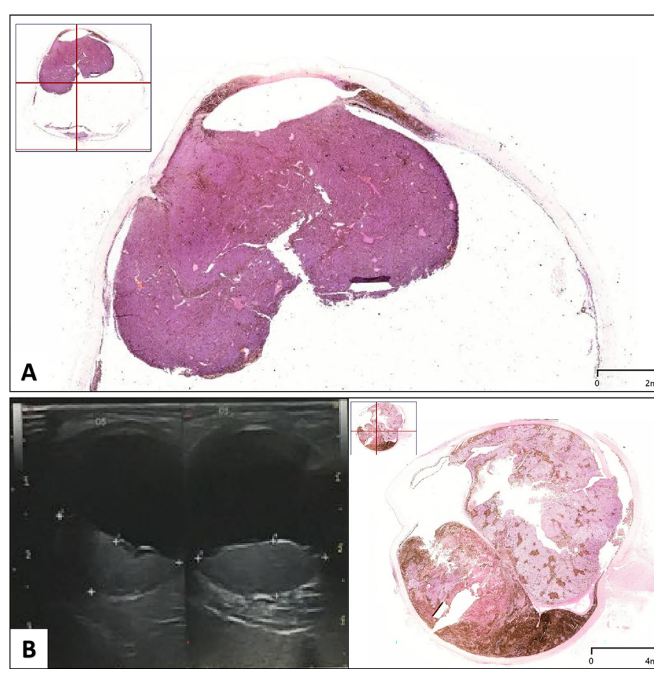

Figure 2 Two tumours had global growth from the secondary enucleation group.

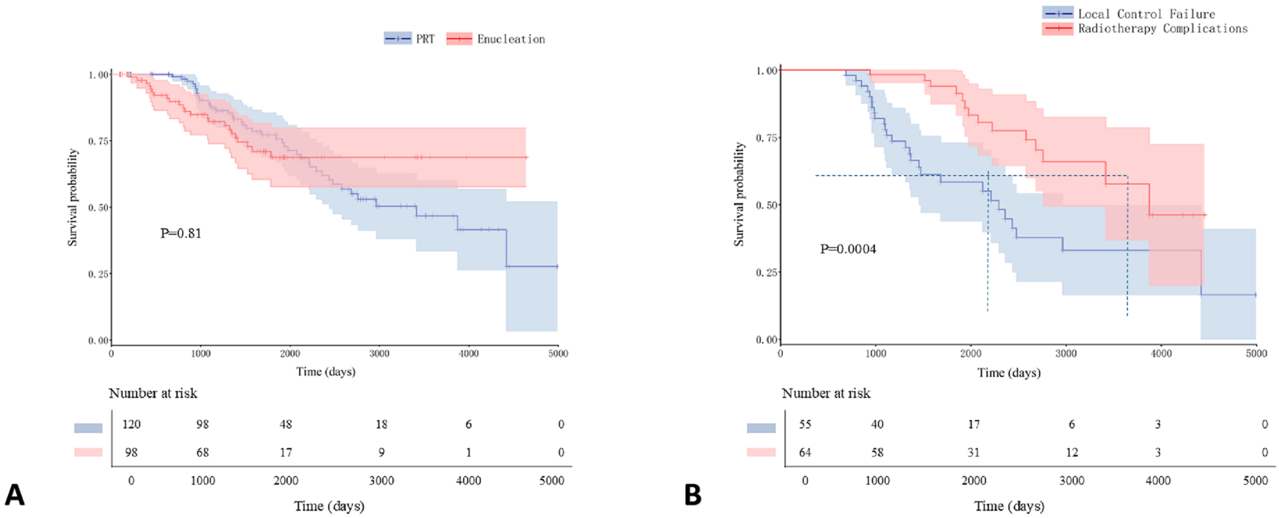

Figure 3 (A) Kaplan–Meier analysis for the PRT group was compared with primary enucleation group with 95% CIs. (B) Kaplan–Meier analysis for secondary enucleation after PRT due to local control failure and radiotherapy complications with 95% CIs. PRT, plaque radiotherapy.

Systemic outcomes and risk factors for metastasis-related death

Online supplemental tables 2 and 3 list the univariate and multivariate factors that predicted tumour metastasis in the PRT and primary enucleation groups. For patients with primary enucleation, univariate analysis revealed that tumour LBD ($p=0.001$), tumour thickness ($p=0.05$), tumour diffuse distribution ($p=0.001$), tumour involvement of the ciliary body ($p<0.001$) and 1 mitosis per 40 HPFs ($p=0.05$) were associated with metastasis-related death. On multivariate analysis, increased age ($p=0.04$), tumour LBD ($p=0.04$), tumour thickness ($p=0.02$) and mitoses≥2 ($p=0.03$) remained significant factors.

For patients with secondary enucleation after PRT, increased age ($p=0.009$), tumour LBD ($p=0.03$), tumour diffuse distribution ($p=0.002$), diffuse pigmented infiltration ($p=0.03$) and mitoses≥2 per 40 HPFs ($p=0.002$) were risk factors for metastasis-related death on univariate analysis. On multivariate analysis, increased age ($p=0.02$), tumour LBD ($p=0.02$), tumour diffuse distribution ($p=0.008$), 1 mitosis per 40 HPFs ($p=0.04$) and mitoses≥2 ($p=0.02$) remained significant.

Table 2 lists the univariate and multivariate factors that predicted tumour metastasis in the local control failure group after PRT. Tumour diffuse distribution ($p=0.04$), moderate pigmented infiltration ($p=0.007$), 10%–50% necrosis ($p=0.02$), diffuse inflammation ($p=0.04$), 1 mitosis per 40 HPFs ($p=0.05$) and mitoses≥2 per 40 HPFs ($p=0.03$) were risk factors for metastasis-related death on univariate analysis. On multivariate analysis, increased age ($p=0.02$), tumour diffuse distribution ($p=0.004$), diffuse inflammation ($p=0.01$), moderate pigmented infiltration ($p=0.005$) and mitoses≥2 per 40 HPFs ($p=0.04$) remained significant.

In addition, six patients with secondary enucleation after PRT, whose histopathological examination showed extensive tumour cell necrosis and could not determine the tumour cell type, still developed metastasis and metastasis-related death.

DISCUSSION

Brachytherapy is the most widely used globe-sparing treatment for UM[17]; however, histopathological data following PRT remain limited. In this study, we compared histopathological characteristics between secondary enucleation specimens after I^{125} brachytherapy and primary enucleation controls to evaluate the effects of radiation on tumour and ocular tissues. Furthermore, we investigated histopathological factors associated with long-term prognosis in patients with UM.

Our analysis revealed distinct histopathological alterations in tumours and surrounding ocular tissues following PRT when compared with primary enucleation specimens. Post-PRT eyes exhibited increased tumour necrosis, tissue fragmentation and inflammatory infiltration, consistent with prior studies.[8 13] These changes reflect typical radiation-induced responses, including vascular damage and necrosis, which are indicative of radiation retinopathy in adjacent retinal tissues.[18] Additionally, eyes enucleated after PRT showed higher rates of scleral and optic nerve invasion, as well as tumour diffuse distribution. It remains unclear whether these features represent inherently aggressive tumours that progressed despite treatment or whether radiation itself contributed to a more invasive phenotype. Previous reports have described both increases and decreases in malignant cell types following radiotherapy.[19] Due to the limited availability of pretreatment fine-needle aspiration biopsy in our cohort, comparative analysis of cellular changes was not feasible. Nevertheless, we found no significant association between pathological tumour type and metastasis or related mortality, aligning with the findings of Char *et al*.[19]

Nevertheless, while 94%–96% of patients with UM achieve local tumour control with brachytherapy at 5 years, a small proportion experience local failure or severe complications ultimately requiring secondary enucleation. Histopathological comparison between eyes enucleated due to local control failure and those removed due to radiotherapy complications revealed several distinctions. Eyes in the local failure group exhibited a higher prevalence of presumed viable tumour tissue, along with greater invasion of the optic disc and ciliary body. In contrast, tumours enucleated due to complications showed significantly lower mitotic activity, consistent with previous reports.[8 13] This may reflect either radiation-induced cytotoxicity or, alternatively, indicate that tumours with inherently higher mitotic activity—potentially associated with more aggressive growth[20]—may be relatively radioresistant, leading to persistent growth and

Table 2 Analysis of risk factors of metastasis-related death in the local control failure group

	Univariate analysis		Multivariate analysis	
Variable	HR (95% CI)	P	HR (95% CI)	P
Age	1.033 (0.994 to 1.075)	0.10	1.055 (1.010 to 1.101)	0.02
LBD	1.081 (0.933 to 1.251)	0.30		
Tumour thickness	1.015 (0.902 to 1.141)	0.81		
Tumour location				
Posterior to equator	1.000		1.000	
At equator	1.648 (0.536 to 5.071)	0.38	1.563 (0.508 to 4.805)	0.44
Anterior to equator	1.647 (0.211 to 12.853)	0.63	3.093 (0.360 to 26.540)	0.30
Ciliary body	0.000 (0.000 to 0.000)	0.99	0.000 (0.000 to 0.000)	0.99
Diffuse distribution	2.856 (1.081 to 7.544)	0.03	5.018 (1.669 to 15.087)	0.004
Involved the optic disc	1.679 (0.734 to 3.843)	0.22		
Involved the ciliary body	1.233 (0.536 to 2.839)	0.62		
Scleral extension	1.263 (0.726 to 2.195)	0.41		
Optic nerve invasion	0.789 (0.334 to 1.866)	0.59		
Iris neovascularisation	0.832 (0.377 to 1.836)	0.65		
Pathological types	1.308 (0.772 to 2.216)	0.32		
Tumour cell necrosis				
None	1.000		1.000	
< 10%	2.066 (0.614 to 6.947)	0.24	2.382 (0.675 to 8.399)	0.18
10%–50%	3.617 (1.196 to 10.942)	0.02	2.459 (0.782 to 7.731)	0.12
50%–90%	1.418 (0.394 to 5.099)	0.59	1.153 (0.302 to 4.400)	0.84
> 90%	3.008 (0.343 to 26.356)	0.32	5.661 (0.575 to 55.729)	0.14
Inflammation				
None or light	1.000		1.000	
Moderate	1.529 (0.576 to 4.061)	0.39	1.298 (0.475 to 3.550)	0.61
Diffuse	2.744 (1.075 to 7.004)	0.04	3.420 (1.294 to 9.040)	0.01
Pigmentation				
None or light	1.000		1.000	
Moderate	5.854 (1.633 to 20.984)	0.007	6.642 (1.770 to 24.924)	0.005
Diffuse	2.742 (0.687 to 10.936)	0.15	3.011 (0.733 to 12.364)	0.13
Tumour vessels prominent	0.917 (0.536 to 1.569)	0.75		
Mitoses				
0	1.000		1.000	
1	3.187 (1.010 to 10.058)	0.05	2.559 (0.741 to 8.841)	0.14
≥2	3.773 (1.160 to 12.272)	0.03	3.853 (1.076 to 13.794)	0.04
Tumour growth	0.757 (0.451 to 1.272)	0.29		

LBD, largest basal diameter.

eventual enucleation. Eyes enucleated owing to radiotherapy-related complications demonstrated more frequent iris neovascularisation, tissue necrosis and structural fragmentation. These findings align with studies on proton beam irradiation, which attribute tumour regression in part to ischaemia.[6 21] Iris neovascularisation is likely a manifestation of radiation-induced ocular ischaemia. Notably, the proportion of iris neovascularisation in our cohort was significantly higher than that reported in previous studies.

Of the 243 patients with UM included in this study, histopathologically confirmed iris neovascularisation was present in 37%. Notably, within the PRT group, 78 eyes (66%) exhibited this complication. Previous studies have reported iris neovascularisation in up to 21% of eyes following I^{125} brachytherapy,[22] while Puusaari *et al* observed incidences of 62% and 60% for iris neovascularisation and NVG, respectively, in a predominantly Caucasian cohort.[23] Obviously, the proportion of iris neovascularisation in the patients we included was significantly higher than in the previous study population. The reason for this difference is unknown, but it may be due to ethnic differences, or it may be related to the dose of radiation. Although optimal management strategies remain undefined, several interventions—including laser photocoagulation, PDT and intravitreal antivascular endothelial growth factor (VEGF) agents—have demonstrated efficacy in treating radiation-induced neovascularisation.[24 25] Panretinal photocoagulation has also shown favourable outcomes for anterior segment neovascularisation.[26] While routine prophylactic anti-VEGF therapy post-PRT remains controversial, our findings underscore the need for further long-term studies to establish effective strategies for mitigating sight-threatening microvascular complications.

The marginal growth after PBT was mentioned by many studies.[27] It might be due to an insufficient radiation dose at the tumorous border following errors in treatment planning or the irradiation delivery.[28] The global growth might be due to tumour radioresistance.[27] Since the radiation dose at the tumour apex is 100 Gy and the irradiation time is determined by the height of the tumour,[12] the occurrence of apical growth might be due to the insufficient radiation dose at the tumour apex caused by the error in the preoperative evaluation of tumour size. The distant growth was rare and not clearly described. It was mainly observed in ciliary tumours notably if there is an iris root involvement.[27] The distant growth might be due to the melanoma cell spreading throughout the anterior chamber or to the extension of the melanoma along the ciliary body (like for 'ring melanoma')[29] or migrating into exudative retinal detachment mentioned by Desjardins *et al*.[28] In our cohort, marginal growth was the most common pattern, followed by global growth, consistent with the existing literature. However, the growth pattern was not identified as a significant risk factor for metastasis-related mortality in univariate analysis.

In our previous study, the 5-year overall survival rate for Chinese patients with UM was 92.7%.[30] In this study, the 5-year metastasis-free survival rate was 76.45% (95% CI 66% to 84%) in patients undergoing secondary enucleation after PRT, compared with 67.95% (95% CI 56% to 78%) in the primary enucleation group. Among secondary enucleation patients, those enucleated due to radiotherapy complications exhibited a markedly higher 5-year metastasis-free survival (91.33%; 95% CI 78% to 97%) than those with local control failure (58.49%; 95% CI 43% to 71%), underscoring that tumour progression after brachytherapy significantly increases metastatic risk and mortality. These findings reaffirm that PRT remains a valuable first-line treatment for eligible patients with UM.

Consistent with the previous studies, we identified larger initial tumour size as a significant risk factor for recurrence. Advanced age was also confirmed as an adverse prognostic factor for survival,[31] possibly reflecting age-related declines in immune surveillance and increased susceptibility to metastatic spread. Notably, higher mitotic activity was a statistically significant predictor of metastasis-related mortality in both primary and secondary enucleation groups—a finding seldom emphasised in earlier research. Furthermore, the role of chronic inflammation in promoting tumour progression, well documented in other solid tumours, may also contribute to UM pathogenesis. Previous analyses of inflammatory cells in irradiated UM revealed that the numbers of tumour-infiltrating T lymphocytes and macrophages varied significantly, and the presence of these infiltrating cells might be related to the pathological type of the tumour cells.[32] Our results further identified diffuse intratumoral inflammation as a statistically significant risk factor for metastasis-related mortality among patients with local control failure after PRT. Notably, six patients in the PRT group developed metastasis and subsequent death despite extensive tumour necrosis on histopathology. This highlights that even with successful local control, patients with UM require long-term systemic surveillance due to the risk of metastatic disease.

Advances in imaging modalities, such as wide-field fundus photography and contrast-enhanced ocular ultrasonography, have improved the clinical detection of UM. Nevertheless, histopathology remains the diagnostic gold standard, particularly for identifying subtle features, such as diffuse growth patterns, early extrascleral extension or optic nerve infiltration—which are challenging to discern with current non-invasive techniques. Early detection of these features is critical for clinical management.

To our knowledge, this study represents the first large-scale histopathological analysis of enucleated UM in an East Asian population. A key strength lies in the correlation between histopathological findings and clinical reasons for enucleation, revealing limitations in clinical assessment—including underestimation of mitotic activity, inflammatory infiltration and anterior segment neovascularisation. By detailing postbrachytherapy morphological changes, we identified microscopic features associated with treatment failure and prognostic outcomes.

Several limitations should be acknowledged. First, the study lacks external validation, which is challenging, given the rarity of UM. Second, its retrospective nature inherently limits data completeness. Third, the sample size, though substantial for this population, may still introduce statistical bias. Finally, growing evidence implicates genetic mutations (eg, in BAP1, SF3B1 and EIF1AX) in UM metastasis risk.[33] Future studies incorporating molecular profiling are needed to elucidate the mechanisms underlying prognostic differences and improve risk stratification.

CONCLUSION

In summary, this retrospective analysis of histopathological features has refined the understanding of indications for secondary enucleation following PRT. We identified a higher prevalence of iris neovascularisation than previously reported based on clinical evaluation alone. Furthermore, microscopic characteristics, particularly mitotic activity, may serve as significant prognostic indicators in UM.

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Ethics approval This study involves human participants. This manuscript does not contain personal and/or medical information about an identifiable individual. The Medical Ethics Committee of the Beijing Tongren Hospital, Capital Medical University peer reviewed and approved the related documents and all study protocols for the collection and use of participants' data. The committee's reference number is TRECKY2018-056. The research followed the tenets of the Declaration of Helsinki. All information is kept confidential, and the patients' basic information will not be made available to other non-researchers or organisations. Participants gave informed consent to participate in the study before taking part.

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